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Research paper

Estimation of Y haplotype frequencies with lower order dependencies

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ABSTRACT

Estimating Y haplotype population frequencies is a demanding task in forensic genetics. Despite the suggestion of various methods, none these have yet reached a level of accuracy and precision that is acceptable to the forensic genetics community. At the basis of this problem is the complex dependency structure between the involved STR loci. Here, we approximate this structure by the use of specific graphical models, namely *t*-cherry junction trees. We apply trees of order three by which dependencies between three STR loci can be taken into account, thereby extending the Chow–Liu method which is restricted to pairwise dependencies. We show that the *t*-cherry tree method outperforms the Chow–Liu method as well as the well-established discrete Laplace method in estimation accuracy.

1. Introduction

The problem of estimating Y haplotype population frequencies from databases has recently attracted much attention. A central part of the problem is that newer multiplexes have an increasing number of Y chromosomal STRs (short tandem repeats). The consequence of this is that the resulting haplotypes are extremely rare in populations, and subsequently are even rarer in databases. The proportion of singletons – that is, haplotypes seen only once in a database – is above 90% for all bar the African populations in databases of several thousand haplotypes [1] in the PowerPlex®Y23 System (PPY23, Promega Corporation, Madison, WI). The 'classical' estimation of haplotype frequencies by the haplotype proportions in a reference database is therefore no longer justifiable. This has led to the development of a variety of more or less refined estimation methods for rare haplotype frequencies comprising Bayesian, non-Bayesian, and combined approaches [2–10].

It should be noted at the outset that in forensic genetics we are usually interested in calculating match probabilities (or likelihood ratios), rather than population frequencies. These quantities are fundamentally different, but one may be used to approximate the other. This difference is often overlooked in forensic science. We recap this argument briefly here. A match probability is the probability of finding someone (else) in the population has haplotype h given we know that a person of interest (POI) has this haplotype. We might write this as

$$\Pr(H_D = h | H_S = h, S \neq D), \tag{1}$$

where H_D is the haplotype of the true donor D, H_S is the haplotype of the POI S, and we are making this calculation under the assumption that the true donor is someone other than the POI. A population frequency, in contrast, is expressed as

$$\Pr(H_D = h). \tag{2}$$

The confusion arises because often the population frequency (Eq. 2) is used to approximate the match probability (Eq. 1) – especially when dealing with autosomal genotypes. This approximation is justified in the autosomal case by arguing that the dependency between *seemingly unrelated* individuals is low. Caliebe and Krawczak [11] argue quite convincingly that approaches that utilise information beyond haplotype databases are needed for Y chromosomal profiles. That being said, estimation of population frequencies inevitability will still be some part of the building blocks of new solutions, and hence we believe that this problem does warrant some attention. In addition, the proposed method provides insight into the complex dependence structure between the different markers used in haplotype determination.

Y chromosomal STR markers, in contrast to the more common autosomal STR markers used in forensics, cannot be assumed to be

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approximately independent. In fact, two opposing forces are in effect. On the one hand, absence of recombination causes dependence between markers. On the other hand opposing this is the mutational process which acts to destroy this linkage. The actual dependence structure between markers is complex [12,13]. A natural way to evaluate how much information is useful in the estimation procedure is to approximate the joint distribution of STR markers through increasing orders of dependency between loci or, in other words, by marginal distributions of an increasing number of markers. We use the terms kth order dependence or marginal distribution to describe a marginal distribution consisting of k markers. This is also sometimes called a k-dimensional marginal distribution. First order dependency then corresponds to independence of markers and the joint distribution of all the markers then is the multiplication of the single marker distributions. Second order dependence takes the pairwise dependency structure into account. The crucial question for haplotype frequency estimation is what degree of dependence, or order, needs to be incorporated into the estimation method to give a good approximation of the complete joint distribution.

Andersen et al. [9] proposed the use of the Chow–Liu algorithm. The resulting tree, for a given data set, from the Chow–Liu algorithm can be shown to be the best approximation allowing for second order dependencies (with respect to the minimal Kullback–Leibler divergence). Not surprisingly, it clearly outperformed first order dependence. The Chow–Liu approximation is a special case of a certain class of graphical models known as *t-cherry junction trees* (or *t*-cherry trees for short). *t*-cherry trees of order k give the best approximation in a certain sense when taking kth order dependencies into account. The Chow–Liu tree is thus the *t*-cherry junction tree of order two ($k = 2$).

In this study we extended the Chow–Liu estimation method of Y haplotype frequencies by incorporating third order dependencies through *t*-cherry trees of order three. We investigated if, and by what degree, estimation accuracy is increased when proceeding from second to third order dependencies by simulating populations and databases for different STR marker sets. For comparison, we also calculated estimates according to the well-established discrete Laplace method [6,14]. In [9] it was shown that the discrete Laplace method and the Chow–Liu method have comparable performance for nine loci but the discrete Laplace has superior performance for 12 loci. Only a small proportion of possible haplotypes could be realised in our simulations of 1.5 million haplotypes because the space of all possible haplotypes is large for nine and 12 loci. This led to intrinsic over- and under-estimation problems for haplotype frequencies. Therefore, to attenuate this problem, we additionally simulated a marker set of four loci.

2. Methods

2.1. Theory of graphical models

Structures of probabilistic dependency can be represented meaningfully by undirected graphs via Markov networks (also called Markov random fields). The vertices in the graph then correspond to random variables while the edges indicate dependence between the variables. A special kind of undirected graph, which we will introduce now, is the junction tree. Each junction tree implies the definition of a probability mass function (pmf) which can be evaluated numerically with relative ease. Junction trees can therefore be used to approximate a certain type of complex dependency structures. In our case, the pmf we want to approximate is the joint pmf of Y STRs.

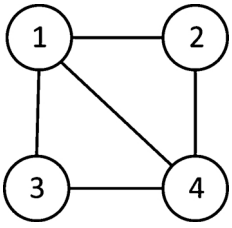


Fig. 1. A triangulated (decomposable) graph.

2.1.1. Basic graph theory

First, let us recall some basic definitions from graph theory [15]. Let $G = (V, E)$ be an *undirected graph* with V the finite set of *vertices* (also called *nodes*) and $E \subset V \times V$ the set of *edges*. A pair of vertices $(v_i, v_j) \in V$ is said to be *connected* if they have an edge $e_{ij} \in E$ between (or joining) them. A *cycle* is a sequence of pairwise connected vertices $v_1, \ldots, v_n \in V$ (i.e. $(v_i, v_{i+1}) \in E$) with an additional edge between v_1 and v_n, and all the vertices are distinct. A cycle is said to possess a *chord* if there is a pair of non-consecutive vertices v_a and v_b in the cycle with $(v_a, v_b) \in E$. A graph is called *triangulated (or chordal)* if every cycle greater than three contains a chord.

For example, the graph in Fig. 1 is a cycle consisting of the vertices {1, 2, 3, 4}. This cycle possess a chord because there is an edge between vertices 1 and 4. This cycle is said to be triangulated because the only cycle of length greater than three is the cycle itself, and it contains a chord. The property of a graph to be triangulated can be shown to be equivalent to the graph being *decomposable*. The term has intuitive appeal, in that if one considers the graph in Fig. 1, then it can be seen that this can be decomposed into two smaller cycles (1, 2, 4) and (1, 3, 4). Decomposability is desirable because if the graph can be decomposed into several smaller components, then these can be treated separately in simpler computations. If the vertices represent random variables, then decomposability is related to conditional stochastic independence. We will explain in what way shortly.

2.1.2. Junction trees

We will explain now this decomposition concept in more detail by using the concept of junction trees. Triangulated graphs and junction trees are closely related because every triangulated graph has a junction tree. Before we can give the definition we need some more graph theory [15]. A *clique* of a graph is a subset of V where all pairs of vertices are connected. A *maximal clique* is a clique that cannot be extended by including an extra vertex. This can be illustrated with an example. Each of the vertices 1, 2, 3, 4 in Fig. 1 all form a clique of order one. None of these are maximal, as the inclusion of any connected vertex is also a clique. That is, if we consider the vertex 1, then the inclusion of either 2, 3, or 4 would make a clique of order two. There are five cliques of order two. This set is every pair of vertices except (2, 3), i.e. {(1, 2), (1, 3), (1, 4), (2, 4), (3, 4)}. Again, none of these cliques is maximal, because the addition of any connected node would yield a clique of order three. There are two cliques of order three. These are (1, 2, 4) and (1, 3, 4). They are both maximal in that the (1, 2, 3, 4) which is obtained by adding either vertex 3 or 2 respectively, is not fully connected because there is no edge between 2 and 3, and is therefore not a clique.

A *path* is a sequence of distinct vertices $v_1, \ldots, v_n$ where consecutive vertices are connected.[2] A subset S of V is called a *separator* of the two subsets A and B of V if every path from A to B intersects S. For example, the set $S = \{1, 4\}$ is a separator of the sets $A = \{1, 2, 4\}$ and $B = \{1, 3, 4\}$.

A *junction tree* (of maximal cliques) $T = (C, \Gamma)$ of a graph G is another representation of a triangulated undirected graph. Its vertices C are all maximal cliques of G. $\Gamma \subset C \times C$ is the set of edges. It, not surprisingly, must be a tree, i.e. all its vertices have to be connected by exactly one path (there are no cycles). Additionally, the *running intersection property* has to be satisfied: For all pairs of cliques C_a, $C_b \in C$ the intersection $C_a \cap C_b$ has to be contained in any vertex on the unique

[2] A cycle is a special case of a path with an extra edge between the starting and ending vertex. A cycle is also a unique sequence of vertices, i.e. no vertex is included more than once.

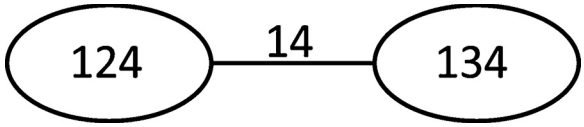


Fig. 2. The junction tree for the graph in Fig. 1.

path from C_a to C_b. If we consider two neighbours C_a and C_b in the junction tree such that $(C_a, C_b) \in \Gamma$, it can be shown that $S = C_a \cap C_b$ is a separator of C_a and C_b in the underlying graph G. The edge between C_a and C_b in the junction tree therefore corresponds to S. A junction tree is thereby determined by the set of maximal cliques C and the set of separators $\mathcal{S}$.

The junction tree for the graph shown in Fig. 1 is shown in Fig. 2. It has two vertices, corresponding to the two maximal cliques in the original graph, and a single edge. It is a tree, in that there are no loops – i.e. no edges from vertices to themselves, and no cycles. It satisfies the running intersection property, in that the intersection of the vertex sets described by the (maximal) cliques $\{1, 4\}$ is the separator between these two cliques.

For more details on graph theory, junction trees and Markov networks see, for example, Cowell et al. [16] and Lauritzen [15].

2.1.3. *t-cherry trees*

We now connect the graph theory with probability theory. Let $X = (X_1, ..., X_n)$ be a discrete random vector with some (unknown) joint pmf P and possibly complex dependencies between the random variables. For $M \subset \{1, ..., n\}$ let X_M be the corresponding joint marginal vector and x_M the projection of $x \in \mathbb{R}^n$. We can then assign a new pmf, P_T, to X for each junction tree T on $G = \{1, ..., n\}$. It is only necessary to know the marginal distributions with respect to P on the set of cliques C and the set of separators $\mathcal{S}$ of the junction tree in order to do this. That is,

$$P_T(X = x) = \frac{\prod_{C \in C} P(X_C = x_C)}{\prod_{S \in \mathcal{S}} P(X_S = x_s)^{\nu_S - 1}} \tag{3}$$

where ν_S is the number of cliques including S. We can see that the cardinality (or size) of the cliques, i.e. the number of vertices (of the original graph) in the cliques, determines the complexity of the numerical calculation of P_T. If k is the maximal number of elements of V contained in a vertex of T, then $k - 1$ is called the *tree width* of this junction tree. A tree width of $k - 1$ therefore means that only marginal distributions of order k have to be regarded and are used to approximate the full pmf of X.

A *t-cherry junction tree* or, for short, *t-cherry tree* is a special type of junction tree. Specifically, a kth *order t-cherry tree* is a junction tree of tree width $k - 1$ where all cliques are of size k and all separators are of size $k - 1$. The term *t-cherry tree* was coined in 2010 by Kovács and Szántai [17], but the concept was developed independently by Malvestuto in 1991 [18]. Malvestuto referred to kth order t-cherry trees as *elementary models of rank* $k - 1$ [18], and later as $k - 1$-*hypertrees* [19]. We adhere to the slightly more expressive name *t-cherry tree*. In [20] it was shown that the number of k-trees on a graph G of n labeled vertices is $\binom{n}{k}(kn - k^2 + 1)^{n-k-2}$. As a kth order t-cherry junction tree on G is a $(k - 1)$-width tree, there are

$$\binom{n}{k-1}((k-1)(n-k+1)+1)^{n-k-1}$$

different kth order t-cherry trees. That means, for example, that there are

$$\binom{12}{3-1}((3-1)(12-3+1)+1)^{12-3-1} = 2,496,308,717,826$$
$$\approx 2.5 \times 10^{12}$$

t-cherry trees for 12 loci where we consider dependencies of order 3. This number increases to 2.78×10^{33} for 23 loci. Such enormous numbers of possibilities mean that exhaustive exploration is impossible, and so we need a smarter method of exploring the space.

Fig. 3 shows (A) a triangulated graph and (B) a corresponding t-cherry tree. Note that the cliques here are of size three, which means that for the approximation we have to take third order dependencies into account.

2.1.4. *Approximating the haplotype distributions*

If we want to obtain a good approximation of an empirical dependency structure, we first have to specify a measure to quantify how well one distribution (here the t-cherry tree distribution) approximates another one (here the full joint distribution of the Y haplotype). An established measure is the *Kullback–Leibler (KL) divergence* which is based on the entropy and therefore is also called *relative entropy*. For two discrete probability distributions P and Q with the same state space E and with P absolutely continuous with respect to Q, the Kullback–Leibler divergence is defined as

$$d_{KL}(P||Q) = \sum_{x \in E} P(x) \log \frac{P(x)}{Q(x)}.$$

Note that the Kullbach–Leibler divergence is not symmetric and therefore not a distance (the triangular inequality does not hold either). It describes the approximation of P by Q. The smaller the value of $d_{\mathrm{KL}}(P||Q)$, the better the approximation. $d_{\mathrm{KL}}(P||Q) = 0$ iff $P = Q$ almost surely. The Kullback–Leibler divergence is closely related to the *mutual information*, I. This can be defined for three discrete random variables with state space E_1, E_2 and E_3 as

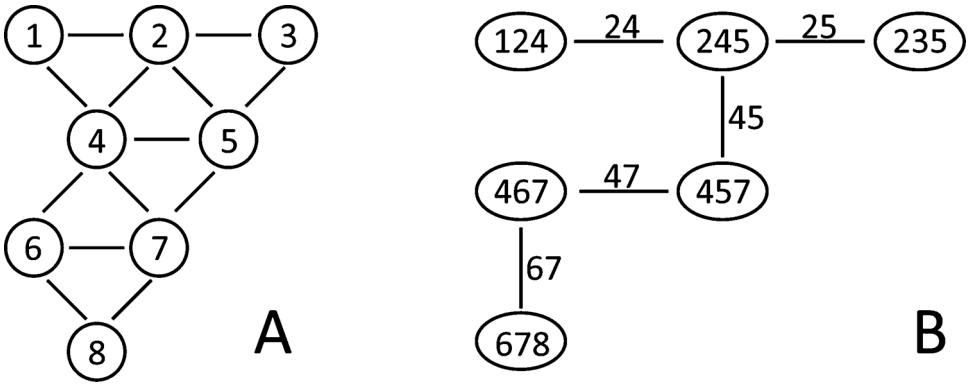


Fig. 3. Example of a triangulated graph and corresponding t-cherry tree. (A) Triangulated graph with eight vertices. The maximal cliques are {{1, 2, 4}, {2, 4, 5}, {2, 3, 5}, {4, 6, 7}, {4, 5, 7}, {6, 7, 8}}. (B) Corresponding t-cherry tree. Here the cliques of the graph of A are the vertices of the t-cherry tree. On the edges of the tree, the separators belonging to the two connected vertices are displayed.

$$I(X_1, X_2, X_3) = \sum_{x_1 \in E_1, x_2 \in E_2, x_3 \in E_3} P(X_1 = x_1, X_2 = x_2, X_3 = x_3) \times \log \frac{P(X_1 = x_1, X_2 = x_2, X_3 = x_3)}{P(X_1 = x_1)P(X_2 = x_2)P(X_3 = x_3)},$$

which can be generalised to more than three random variables in several ways. For three variables we thus have

$$d_{\mathrm{KL}}(P_{(X_1,X_2,X_3)} || P_{X_1} \otimes P_{X_2} \otimes P_{X_3}) = I(X_1, X_2, X_3),$$

where $P_{X_1} \otimes P_{X_2} \otimes P_{X_3}$ is the product measure. Mutual information has been successfully used as the means of assessment of dependencies between Y STRs in [9,12,21].

It is necessary to restrict the (tree) width of the junction tree in order to get a manageable approximation of a distribution by a junction tree distribution. The task is, therefore, to find a junction tree with fixed tree width which minimises the Kullback–Leibler divergence with respect to the full joint distribution of the Y haplotype. The significance of t-cherry trees is that within the class of junction trees with tree width $k-1$, the minimal Kullback–Leiber divergence is achieved by a k order t-cherry tree [18,22]. As described earlier, all cliques are of size k and all separators of size $k-1$ in a k order t-cherry tree, and the approximation is as dense as possible for that k (it is not possible to add another edge without introducing a dependency of higher order than k), which provides intuition why this class of models has such desirable theoretical properties. The distribution of Y haplotypes was approximated in Andersen et al. [9] by the Chow–Liu tree which is a t-cherry tree of second order and allows for second order dependencies between Y STR loci. In this study we extend this to t-cherry trees of third order, thus incorporating third order dependencies.

2.1.5. Constructing t-cherry trees in this study

Constructing the optimal t-cherry tree is an NP hard problem [23,24]. Therefore, we designed, and implemented, two versions to construct t-cherry trees of third order as approximations to the optimal solution. We implemented the algorithms in the open-source R package `tcherry` available at https://github.com/nvihrs14/tcherry [25]. The first version (via the function `tcherry_CL()` in `tcherry`) uses the Chow–Liu tree as a starting point, and seeks to extend it. This version is similar to Algorithm 2 proposed in [17]. The second version uses the triplet of random variables with the highest mutual information as the initial clique and adds successively cliques until all vertices are incorporated (via the function `tcherry_step()` in `tcherry`). This version is similar to algorithms proposed in [26,18]. Since we want to construct t-cherry trees of third order we have to estimate the underlying third order marginal distributions for both versions. We did this simply by using the empirical fractions from the three way contingency tables with support being the Cartesian product of the observed alleles at each locus.

In order to find the junction tree that minimises the Kullback–Leibler divergence with respect to the underlying true distribution of the joint Y haplotype it is sufficient to maximise the weight of the junction tree. We will now state the weight functions used by the two versions. Recall that $\mathcal{C}$ is the set of maximal cliques, $\mathcal{S}$ the set of separators, and ν_S is the number of cliques including S (such that $\nu_S - 1$ is the number of times the separator S appears in the junction tree). Version 1 (implemented in `tcherry_CL()`) attempts in a greedy way to maximise the weight

$$\sum_{\{X_i,X_j,X_k\} \in \mathcal{C}} I(X_i, X_j, X_k).$$

Version 2 (implemented in `tcherry_step()`) attempts in a greedy way to maximise the weight

$$\sum_{\{X_i,X_j,X_k\} \in \mathcal{C}} I(X_i, X_j, X_k) - \sum_{\{X_i,X_j\} = S \in \mathcal{S}} (\nu_S - 1) \cdot I(X_i, X_j),$$

which is the correct expression to maximise considering the equation for the Kullback–Leibler divergence between the true probability distribution and the one approximated by a junction tree [17].

Version 1, constructed from the Chow–Liu tree
Input: Data with realisations of the variables

1. Construct the Chow–Liu tree from the data.
2. Calculate the estimated mutual information for all combinations of three variables (triples). Order the triples in descending order with respect to decreasing mutual information.
3. Adding the first cherry:
   (a) Consider the first triple and the associated edges from the Chow–Liu tree.
   (b) If only one edge needs to be added to make the set of three vertices complete, then add this edge and let this clique be the first cherry.
   (c) Else, consider the next triple.
   (d) Repeat (b)–(c), until a cherry is accepted.
   (e) Remove the accepted triple from the list of triples.
4. Adding the rest of the vertices:
   (a) Consider the first triple in the current list of triples and the associated edges from the Chow–Liu tree.
   (b) If two of the vertices in the triple are already in the t-cherry tree and if only one edge needs to be added to make the set of vertices complete, add this edge and let this clique be the next cherry.
   (c) Else, consider the next triple.
   (d) Repeat (b)–(c), until a cherry is accepted.
   (e) Remove the accepted triple from the list of triples.
   (f) Remove all triples from the list of triples where all three variables are already in the t-cherry tree (although in different cherries).
   (g) Repeat (a)–(f) until all vertices are in the t-cherry tree.

Output: Adjacency matrix of vertices and edges for the t-cherry tree

Version 2, constructed directly from the data
Input: Data with realisations of the variables

1. Calculate the estimated mutual information (I) for all combinations of three variables (triples).
2. Let the triple with highest mutual information be the first cherry.
3. Adding the rest of the vertices:
   (a) Calculate for all vertices v_k which are not yet in the t-cherry tree and all edges (v_i, v_j) in the t-cherry tree $w(v_i, v_j, v_k) = I(v_i, v_j, v_k) - I(v_i, v_j)$.
   (b) For the highest value $w(v_i, v_j, v_k)$ in (a), make $\{v_i, v_j, v_k\}$ complete and let this clique be the next cherry.
   (c) Repeat (a)–(b) until all vertices are in the t-cherry tree.

Output: Adjacency matrix of vertices and edges for the t-cherry tree.

Fig. 4 shows exemplar t-cherry trees for 12 forensic STR markers constructed with versions 1 and 2 using data from the Danish reference database [27]. Although the two constructed t-cherry trees are different in this example, in our simulations the two versions yielded mostly identical junction trees.

2.2. Simulations

Simulation of reference haplotype populations and databases were performed as described in [9]. We recapitulate here the basic principles. We first simulated 20 genealogies of 1.5 million individuals each under different population genetic assumptions. The scenarios varied with respect to population growth and reproductive success (Table 1). The mutation process of three different marker sets was superimposed to each the 20 genealogies. These marker sets consisted of 4, 9, and 12 loci and were labelled 4-LMS (four loci marker set), 9-LMS, and 12-LMS respectively. The 9-LMS and 12-LMS correspond to the so-called *minimal* and PowerPlex Y marker sets. These were also previously used

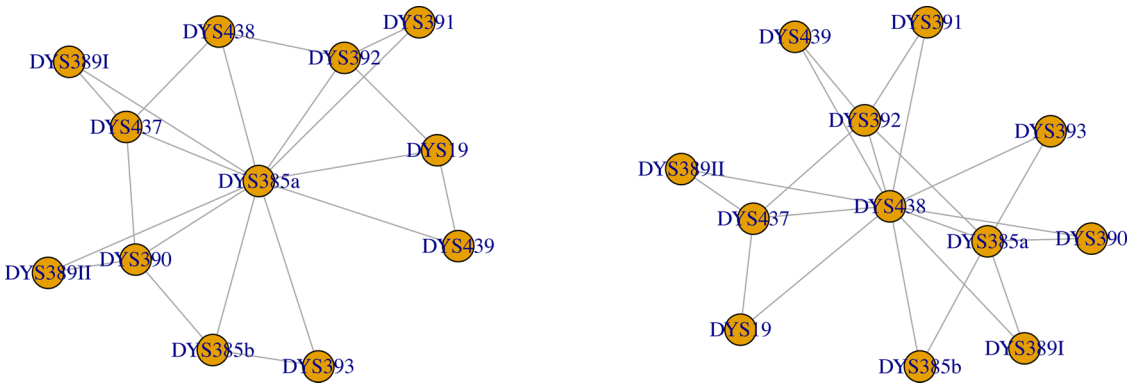


Fig. 4. Example of *t*-cherry trees constructed with versions 1 and 2. The *t*-cherry trees were constructed using the Danish reference database [27]. (A) *t*-cherry tree constructed according to Version 1 (from the Chow–Liu tree). (B) *t*-cherry tree constructed according to Version 2 (direct construction).

Table 1
Scenarios for genealogy simulations. α is the growth rate per generation in %, *VRS* is variance in reproductive success between males, and n is number of simulations.

α	*VRS*	n
0	None	5
0	High	5
2	None	5
2	High	5

Table 2
Markers in the 4-LMS. μ is mutation rate according to YHRD.org release 55 [30].

Marker	Ladder size	μ
DYS389I	9	0.00293
DYS437	10	0.00122
DYS391	12	0.00245
DYS19	12	0.00224

in [9]. The 4-LMS was newly implemented. The aim was to avoid the problem of extremely low haplotype frequencies for this set and to cover the space of all possible haplotypes as best as we could. Therefore, the markers of the 4-LMS were selected from the complete 12-LMS in such a way as to generate a small set of all possible haplotypes, each with a reasonably high haplotype frequency. The selection criteria were primarily a short allelic ladder and secondly, if the primary criterion was not unique, a large mutation rate. This resulted in the set of markers shown in Table 2. The number of possible different haplotypes is then $9 \times 10 \times 12 \times 12 = 12{,}960$. A symmetric step-wise mutation model with reflecting boundaries was used. Each marker set was simulated five times for each of the 20 genealogies resulting in 100 reference haplotype populations per marker set and 300 reference haplotype populations altogether.

All simulations and analyses (see next section) were performed in R [28]. The simulations also made use of the R package `malan` [29].

2.3. Frequency estimation

Fifty (50) databases of size 500 and of size 1000 each were drawn from all 300 reference haplotype populations. An additional haplotype h was drawn from the reference population for each database for haplotype frequency estimation. The database, augmented by h, was then used as the basis for frequency estimation of h by various methods (see below). Estimated frequencies were compared to the simulated reference haplotype population frequencies to evaluate the estimation accuracy of the methods. This was done for short by the mean squared error (MSE) and bias on the $\log_{10}$ scale. In detail, if n haplotypes $h_1, \ldots, h_n$ are to be estimated, let p_i and $\hat{p}_i$ denote the simulated and estimated frequencies of h_i, respectively for $i \in 1, \ldots, n$. Note that we estimated $n = 15{,}000$ haplotypes for each database size (500 or 1000). The MSE and the bias are then defined as

$$\mathrm{MSE}(\hat{p}) = \frac{1}{n}\sum_{i=1}^{n} (\log_{10}(\hat{p}_i) - \log_{10}(p_i))^2,$$

$$\mathrm{bias}(\hat{p}) = \frac{1}{n}\sum_{i=1}^{n} (\log_{10}(\hat{p}_i) - \log_{10}(p_i))$$

so a positive bias corresponds to an over-estimate, and correspondingly a negative bias to an under-estimate. Ideally both the MSE and bias should be as close to zero as possible.

The following estimation methods were used in this study:

1. *Independence method*: Allele frequencies were estimated by the classical maximum likelihood estimator, dividing the count of a certain allele by the total number of alleles in the database (augmented by the additional haplotype). Haplotype frequency estimation was carried out by simply multiplying the appropriate allele frequency estimates.
2. *Chow–Liu method*: The Chow–Liu tree was computed using the Chow–Liu algorithm [31]. Probability distributions of single loci, and joint probability distributions of two loci, were estimated from the one- and two-way contingency tables of the augmented databases. Again, the simple maximum likelihood estimators were used. The R packages `bnlearn` [32] and `gRain` [33] were applied.
3. *t-cherry tree method*: The two versions, described above, were used to calculate two *t*-cherry tree estimators for each drawn haplotype. We used the R package `tcherry` [25] (available at https://github.com/nvihrs14/tcherry) for finding the best structure (cliques and separators) and then `gRain` [33] to construct the undirected graphical probability model and perform calculations. We applied a smoothing parameter 10^{-3}, i.e. 10^{-3} was added to all cells to avoid zero entries in the contingency tables. An appropriate value for the smooth parameter must be chosen by the user in any application of this method.
4. *Discrete Laplace method*: We used a maximum of 10 and 15 clusters for databases of size 500 and 1000, respectively for the discrete Laplace method [6] using the R package `disclapmix` [34,14]. The frequency of the haplotype in question was estimated for models with from 1 to 10 clusters for databases of size 500 and from 1 to 15 clusters for databases of size 1000. The estimate was chosen in a conservative way as the one with the highest probability (out of the 10 or 15 models). That is, if h is the haplotype in question, c is the number of clusters, and $\mathcal{M}_c$ is the model with c clusters, then we choose

$$\operatorname*{argmax}_{c \in C} \Pr(h|\mathcal{M}_c)$$

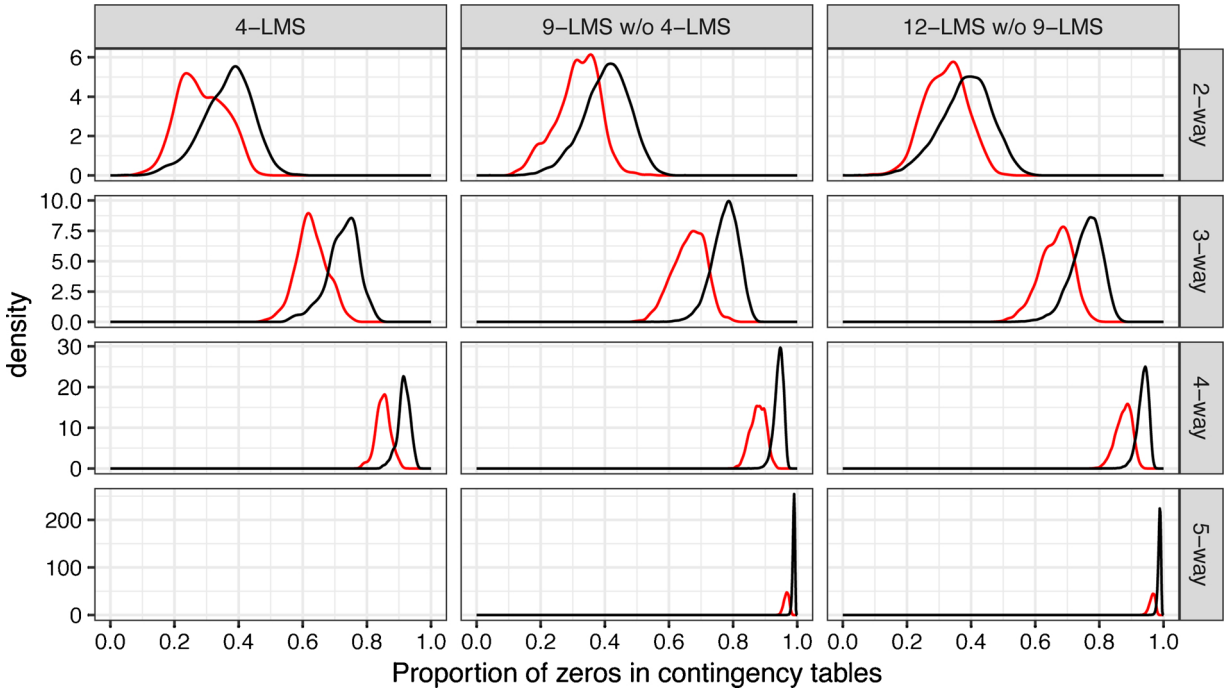


Fig. 5. Distribution of proportion of zeros in contingency cells in the simulated populations and in databases of size 500. Red line: simulated populations, black line: databases of size 500, 4-LMS: cells for all combination of markers in the 4-LMS were regarded, 9-LMS: cells for combination of markers were regarded which had not been included in the 4-LMS calculation, 12-LMS: cells for combination of markers were regarded which had not been included in the 4-LMS or the 9-LMS calculations. For simulated populations, the distributions were calculated from altogether 20 populations per line (one population simulation for each of the 20 genealogies (cp. Table 1)). For databases of size 500, 10 databases for each of the 20 simulated populations were used resulting in 200 databases per line. For the 4-LMS, five way contingency tables do not exist. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Singleton and haplotype proportions in populations and databases.

type	n loci	prop(singleton)	prop(haplotype)
pop	4	3.47×10^{-5}–7.40×10^{-5}	0.070–0.102
pop	9	0.004–0.007	1.20×10^{-6}–2.32×10^{-6}
pop	12	0.008–0.014	1.01×10^{-9}–1.65×10^{-9}
500	4	0.209–0.219	–
500	9	0.885–0.945	–
500	12	0.951–0.976	–
1000	4	0.125–0.132	–
1000	9	0.802–0.897	–
1000	12	0.909–0.952	–

type: type for which proportions were calculated, either pop (population), 500 (databases of size 500) or 1000 (databases of size 1000), n loci: number of loci, prop: For simulated populations, the proportions (of singletons or haplotypes) are calculated from altogether 20 populations per marker set (one population simulation for each of the 20 genealogies (cp. Table 1)). Shown are the minimal and maximal values of these twenty proportions. For databases, all 100 databases per markerset were used. prop(singleton): singleton proportions, prop (haplotype): proportion of realised haplotypes with respect to the number of all possible haplotypes.

as the estimated probability where $C = \{1, \ldots, 10\}$ if the database size is 500, and $C = \{1, \ldots, 15\}$ if the database size is 1000.

Kernel density estimation for the proportion of zeros in contingency tables in simulated populations and databases (Fig. 5) was performed with the function `density` in R for a Gaussian smoothing kernel with default settings (for details see legend to Fig. 5).

3. Results

The main aim of this study was to compare different Y haplotype frequency estimation methods. For this to be meaningful, it is necessary to have some frequency values as reference for accuracy evaluation. Because haplotype frequencies for a reasonable number of loci tend to be low [1], it is not possible to obtain these reference frequencies from present databases, which at most contain some thousands of haplotypes per population. Therefore, we simulated large haplotype populations of 1.5 million haplotypes. The frequencies in these populations serve as reference frequencies and are called simulated frequencies in the following. Databases of size 500 and 1000 were then repeatedly drawn from the populations, resulting in 15,000 databases for each size. Frequency estimates were based on the databases and compared to the simulated frequencies.

We regard here three different marker sets. The first two marker sets consist of nine (9-LMS, nine loci marker set) and twelve loci (12-LMS) and correspond to the frequently used 'minimal' and PowerPlex Y kits. To investigate the special issues of estimating rare haplotypes, we also simulated a marker set with only four loci (4-LMS) for comparison which resulted in much higher haplotype frequencies.

3.1. Simulated population structure

We simulated populations of 1.5 million haplotypes. Table 3 shows the proportion of singletons and the proportion of realised haplotypes (compared to the number of all possible haplotypes in the populations). The proportion of singletons is very low and, as expected, increases with an increasing number of loci. It reaches a maximal value of only 1.4% in the 12-LMS. The proportion of realised haplotypes is between 7% and 10% for the 4-LMS. Much lower values of around 2.0×10^{-4}% and around 1.5×10^{-7}% hold for the 9-LMS and 12-LMS, respectively. The reason for these low values is that the simulated haplotypes share some relationship with some of the other haplotypes in the population. Therefore, clusters of similar haplotypes exist which causes a low number of singletons and on the other hand prevents that a large number of different haplotypes are generated.

For the Chow–Liu and the *t*-cherry tree estimation methods, the

second and higher order dependencies between loci are relevant, respectively. Fig. 5 (red lines) shows the distribution of the proportion of zeros in the cells of two to five way contingency tables of simulated populations. A zero means that the empirical marginal distribution of the considered loci has a probability of zero for this special combination of alleles in the simulated population. Only alleles observed in the population were regarded. It can be seen that these distributions do not differ much between the three marker sets. Because the 4-LMS was chosen in order to generate relatively large haplotype frequencies, its proportions of zeros are somewhat smaller. Large differences can, however, be observed for different orders of dependence. The distribution of the proportion of zeros in two way contingency tables has a support between 0.1 and 0.5, with a maximum of the density around 0.3. The distribution is shifted to considerably higher values for third order dependencies with a support between 0.5 and 0.8 and a maximum around 0.65. Looking at fourth order marginal distributions, the support lies between 0.8 and 0.95 (maximum around 0.85). Finally, the maximum of the densities is as high as around 0.97 for fifth order dependence.

3.2. Database structure

The mean singleton proportions in the databases are given in Table 3. As could be expected, the singleton proportions increased with the number of loci and decreased with database sizes. Therefore the smallest singleton proportions were obtained for database sizes of 1000 for the 4-LMS with values around 13% whereas the largest values around 96% were found for database sizes of 500 for the 12-LMS. For more detailed information see Fig. A1.

The Chow–Liu and the *t*-cherry tree estimation methods approximate the joint pmf of all haplotype loci by second and higher order dependencies, respectively. These are estimated from the respective contingency tables of the databases. In Fig. 5 (black line) the distribution of the proportion of zeros in contingency cells in databases of size 500 is displayed. As for populations (red line), we again see similar distributions across marker sets but a strong shift of the support towards higher values for increasing order of dependence. Not surprisingly, the proportions of zeros for databases show higher values than those for populations. The support is shifted by roughly 0.10 to larger values for two to four way contingency tables.

3.3. Frequency estimation

3.3.1. Comparison of estimation methods

The methods independence, Chow–Liu, *t*-cherry tree and discrete Laplace were applied to estimate the simulated haplotype population frequencies from the generated databases. Fig. 6 and Table 4 display the results for the comparison of the performance of estimation methods. Whereas the independence method does not take any dependencies between loci into account, the Chow–Liu and the *t*-cherry tree methods, as applied in this study, rely on second and third order dependencies, respectively. It can be seen that the more dependency is included in the model, the better is the estimation of the simulated frequencies for the 9-LMS and the 12-LMS. The mean squared error (MSE) and the bias decrease from independence over Chow–Liu to *t*-cherry tree methods. The 4-LMS shows a different behaviour which will be analysed in Section 3.3.3. In the 12-LMS the MSE is considerably lower for the second *t*-cherry tree version than for the first *t*-cherry tree version. This is due to the fact that for the first version the *t*-cherry tree is constructed as an extension of the Chow–Liu tree whereas for the second version the higher order junction tree is constructed from scratch, and because the two versions optimises different weight functions (see Section 2.1.5). Obviously, the second method admits more possible trees to be considered. It can be observed that the discrete Laplace method had a lower MSE than the *t*-cherry trees, and that most of the time it had a lower MSE than the Chow–Liu method. Again, the 4-LMS is an exception.

3.3.2. True vs. simulated vs. estimated frequencies

The set of possible haplotypes for common marker sets like the 9-LMS and the 12-LMS is huge and therefore most haplotypes have very low population frequencies. This is the reason why singleton proportions in databases tend to be very high, and this can be seen in our simulated databases. Moreover, even in our simulated populations of 1.5 million haplotypes, the proportion of different haplotypes realised (or observed) out of all possible haplotypes is very low for the 9-LMS and the 12-LMS (Table 3). This means that we have to differentiate between 'true' haplotype frequencies p_t which would be realised in an infinitely large population and the simulated haplotype frequencies p_s for our population simulations. Note that these two might differ substantially. Because most haplotypes are rare, most of them are not simulated in our populations and thus $p_t > p_s = 0$. On the other hand given a specific simulated population, the realised simulated haplotypes just 'got lucky'. For another simulation, other haplotypes will arise. Therefore, for the simulated haplotypes $p_s > p_t$. In this study we compare the estimated frequencies $\hat{p}$ to the simulated frequencies p_s. We have to bear in mind that a good approximation of p_s by $\hat{p}$ does not mean that we have a good approximation of p_t by $\hat{p}$ because $p_t \neq p_s$.

All estimation methods have a support in a set of possible haplotypes (often much) larger than the set of different haplotypes in a simulated population for the 9-LMS and the 12-LMS because they assign a positive probability to haplotypes not realised in the simulated population. For independence, Chow–Liu and *t*-cherry methods this is achieved by using the marginal probability distributions according to (Eq. 3). As can be seen from Fig. 5, the proportion of zeros in the contingency tables increases with the order of dependency considered and therefore the support is largest for the independence method, medium for the Chow–Liu method and lowest for the *t*-cherry tree method, although still larger than the simulated support. Because less probability mass remains for realised haplotypes and because for our estimation evaluation we only regarded realised haplotypes, the consequence is that in general the estimated frequencies underestimated the simulated frequencies as can be seen by the negative bias in Table 4 for the 9-LMS and the 12-LMS. This phenomenon is the more important the more markers are considered and the less dependency is included in the model. The discrete Laplace method is based on another model and supports the Cartesian product of the complete set of integers, $\mathbb{Z}$, as possible haplotypes. Therefore, it also shows some bias. It is not necessarily a disadvantage that the estimation methods assign positive probability to a number of haplotypes with $p_s = 0$. Indeed, because these haplotypes have $p_t \neq 0$, it means that they could turn up in another simulation and $\hat{p}$ might be a good, unbiased estimator for p_t.

In the 4-LMS, because of higher haplotype frequencies, the set of possible haplotypes is far better explored than for the 9-LMS and the 12-LMS. Therefore, we no longer have predominantly underestimation but indeed have rather a slight overestimation which will be explained in the next section.

3.3.3. Augmenting the database

In the results analysed so far, the database was augmented by the additionally drawn haplotype whose frequency was to be estimated. This is motivated on the one hand by a Bayesian viewpoint [8]: Since the haplotype was drawn from the population it seems reasonable to take that into account for the estimation procedure. The augmentation assures that the frequency estimate will always be larger than zero which coincides with intuition in that one would not want to give an observed haplotype a frequency of zero. Furthermore, augmentation of the database eases the numerical calculations. For example, if the haplotype in question contains an allele not observed in the database

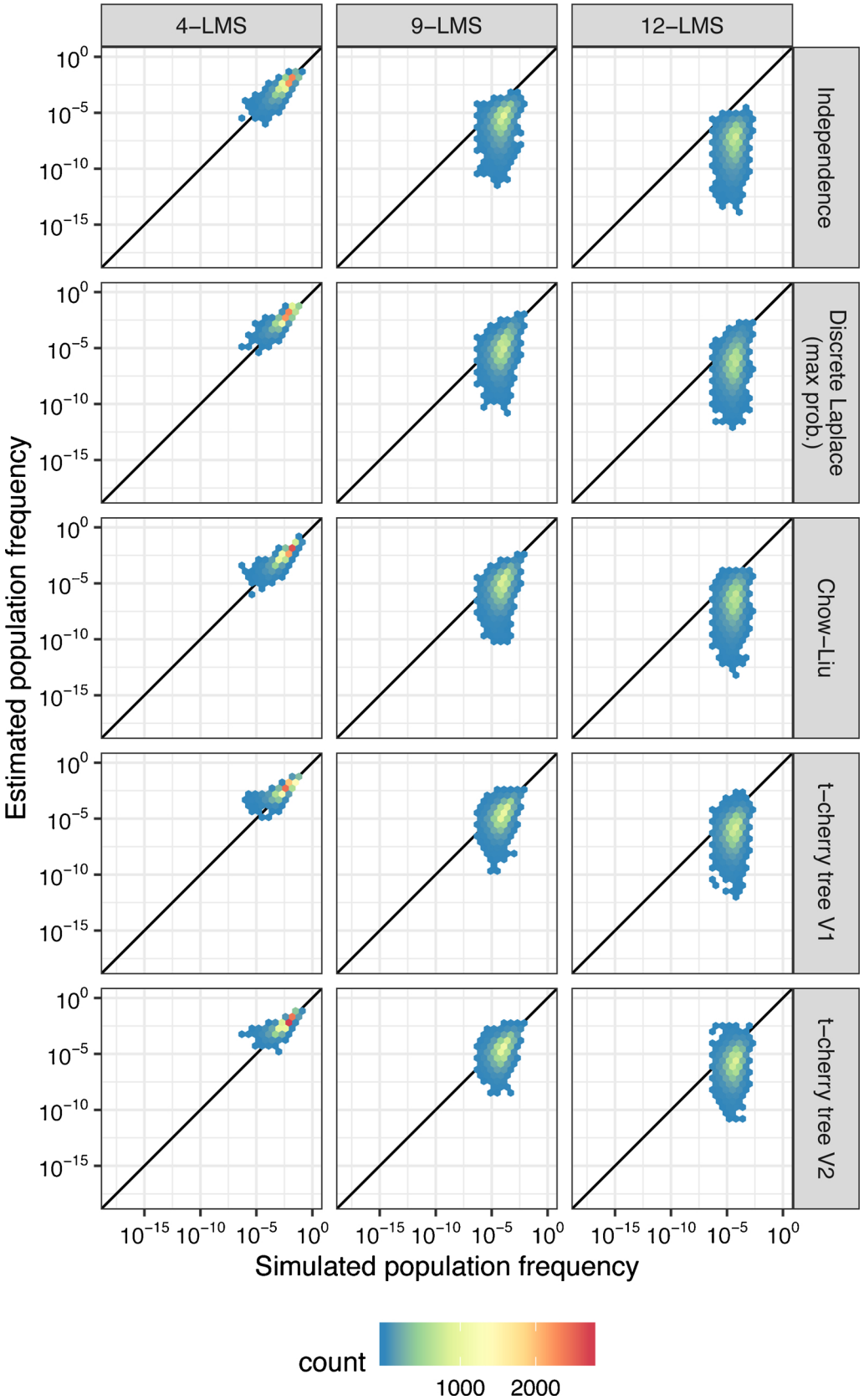


Fig. 6. Comparison of the estimators to the simulated frequencies. The superimposed identity line ($y = x$) is shown. For this figure the results from databases of size 500 and 1000 were merged. V1: first version for the *t*-cherry tree construction (from the Chow–Liu tree), V2: second version for the *t*-cherry tree construction (direct construction).

Table 4
Mean squared error and bias between frequency estimates and simulated frequencies.

DB size	Estimator	4–LMS		9–LMS		12–LMS	
		MSE	Bias	MSE	Bias	MSE	Bias
500	Independence	0.11	−0.11	4.22	−1.72	12.66	−3.30
500	Chow–Liu	0.06	0.00	2.22	−1.13	6.76	−2.26
500	*t*-cherry tree V1	0.11	0.11	1.00	−0.55	3.79	−1.61
500	*t*-cherry tree V2	0.11	0.11	0.91	−0.47	2.96	−1.32
500	Discrete Laplace	0.10	0.15	2.25	−1.06	5.64	−1.96
1,000	Independence	0.12	−0.12	4.29	−1.75	12.71	−3.31
1,000	Chow–Liu	0.05	−0.02	2.51	−1.24	7.40	−2.37
1,000	*t*-cherry tree V1	0.07	0.05	1.34	−0.79	4.83	−1.87
1,000	*t*-cherry tree V2	0.07	0.05	1.21	−0.73	3.98	−1.65
1,000	Discrete Laplace	0.09	0.14	2.24	−1.09	5.79	−2.02

The mean squared errors (MSEs) and biases are given on the $\log_{10}$ scale. A positive (negative) sign of the bias shows on overestimation (underestimation) of the simulated frequencies.
DB: database, V1: first version for the *t*-cherry tree construction (from the Chow–Liu tree), V2: second version for the *t*-cherry tree construction (direct construction).

that would not pose a problem. Augmenting the database is widely used. The simplest example is haplotype frequency estimation by the database fraction where usually the augmented database is the estimation basis.

From a frequentist view, when augmenting the database, we introduce bias because we no longer have independent data. The haplotype to be estimated is given additional weight and it is overrepresented in the database and thus overestimated. This is especially pronounced for rare haplotypes and small databases. We will examine this phenomenon for the 4-LMS. Because of the relatively simple dependency structure the estimation models can approximate the simulated frequencies fairly well. The more complicated a model is, the larger is the effect of the inclusion of the haplotype in question. Therefore, the overestimate is larger for the *t*-cherry tree methods than for the Chow–Liu method which in turn is larger than for the independence method. For the 9-LMS and the 12-LMS the overestimation effect of the augmentation is on the one hand superimposed by the underestimation effect illustrated in the previous subsection. Furthermore, the dependency structures are more complex and cannot be captured as efficiently by the methods.

The results between augmented and non-augmented estimation methods are compared in Fig. 7 and Table 5. If the database is no longer augmented by the haplotype to be estimated, then it is possible for estimated haplotype frequencies to be zero (in Table 5 the number of zero estimates is given as n_0). All estimated probabilities below 10^{-16} were replaced by 10^{-16} to avoid numerical problems—relating to taking the logarithm of zero – when calculating bias and MSE. Therefore clusters of estimated frequencies at this value can be seen for some methods in Fig. 7. Table 5 shows that indeed the bias is lower for the non-augmented than for the augmented estimators. For most methods and database sizes, this bias is now reverted. We now have an underestimation instead of an overestimation and the absolute value is much larger. Why is that the case? The reason is probably that we use the logarithmic values in the calculation of the bias and MSE. It is well recognised that logarithmic values better represent the weight of evidence. A difference between estimated and simulated frequencies has a much greater impact for small frequencies than for larger ones because of the slope of the logarithmic function. If we do not augment the database, then frequencies may be estimated as zero (or 10^{-16}) for haplotypes which are present in the simulated populations thus strongly influencing bias and MSE. This can be clearly seen for the database fraction where the bias shows a strong underestimation when n_0 is large. To a lesser degree this applies also to the independence, the Chow–Liu and the *t*-cherry tree methods. We have an additional parameter for the *t*-cherry trees which is the smoothing parameter. This is a small number added to each cell of the contingency tables. While this parameter has virtually no influence if augmented databases are used, a higher smoothing parameter yields better estimation performance for non-augmented databases because very small estimated frequencies are avoided. The discrete Laplace method is based upon a completely different estimation approach and gives very similar values for augmented as for non-augmented databases.

4. Discussion

In this study, we suggested the use of graphical models for the estimation of Y haplotype population frequencies from databases. Particularly, we proposed the *t*-cherry tree method which takes into account third order dependencies between Y STR loci. This is an extension of the Chow–Liu method, which is based upon second order dependencies of Y STR loci. The *t*-cherry estimation method showed a considerable improvement over the Chow–Liu method indicating that second order dependencies are not sufficient to approximate the dependency structure of Y STR loci. Additionally, it outperformed the well-established discrete Laplace method.

We restricted ourselves only to *t*-cherry trees for third order dependencies which require three way tables of the databases. These three way tables had a high proportion of zeros. Nevertheless, the *t*-cherry estimation method performed well from which we can conjecture that this is no impediment for the application of this method. The consideration of fourth or higher order dependencies seems not to be meaningful regarding the very high proportion of zeros found in the four way contingency tables for databases. Interestingly, the proportions of zeros in databases were only somewhat higher than those detected in the much larger underlying populations. This means that the limited information about higher order dependencies is not due to small database sizes but is an inherent problem of the genealogies of the populations. This is connected to the fact that even for populations of 1.5 million haplotypes only a small proportion of all possible haplotypes are realised in one population simulation due to clusters of haplotypes caused by close relatedness.

The approximation of discrete probability distributions of n variables by junction trees of order k is not an easy task. Indeed, obtaining the tree of order k with the lowest Kullback–Leibler divergence with respect to the empirical haplotype distribution was proved to be NP hard for general k, e.g. [23,24]. The solutions are trivial for $k = 1$ and $k = n$. Another exception is the Chow–Liu tree ($k = 2$) which can be constructed easily with the help of the mutual information. This is, however, no longer the case when we regard *t*-cherry trees of order three or higher. Here, it is necessary to search the whole set of *t*-cherry trees to obtain the optimal tree with the lowest Kullback–Leibler divergence. A number of practical work-arounds have been proposed. In [17,35], the *t*-cherry tree is calculated in a forward approach by extending the Chow–Liu tree. A backwards greedy approach is introduced in [19]. Szántai and Kovácz [26] and Malvestuto [18] proposed two similar greedy algorithms which compute the *t*-cherry tree from scratch by starting with one cherry and then adding the next cherries successively according to a weight function. The two algorithms differ with respect to the weights which are assigned to the cliques. The algorithm of Szántai and Kovácz [26] reaches the optimal solution under certain assumptions on the probability distribution to approximate. In this study, two greedy approaches for approximating the optimal *t*-cherry tree were implemented. The first calculated the tree from the Chow–Liu tree similar to [17] and the second constructed the tree similar to [26,18]. Overall, the two *t*-cherry tree methods performed similar in estimating haplotype frequencies. The second version, however, resulted in somewhat better values which is understandable because it is not restricted to be an extension of the Chow–Liu tree and it also uses another weight function (weights from both cliques and separators and

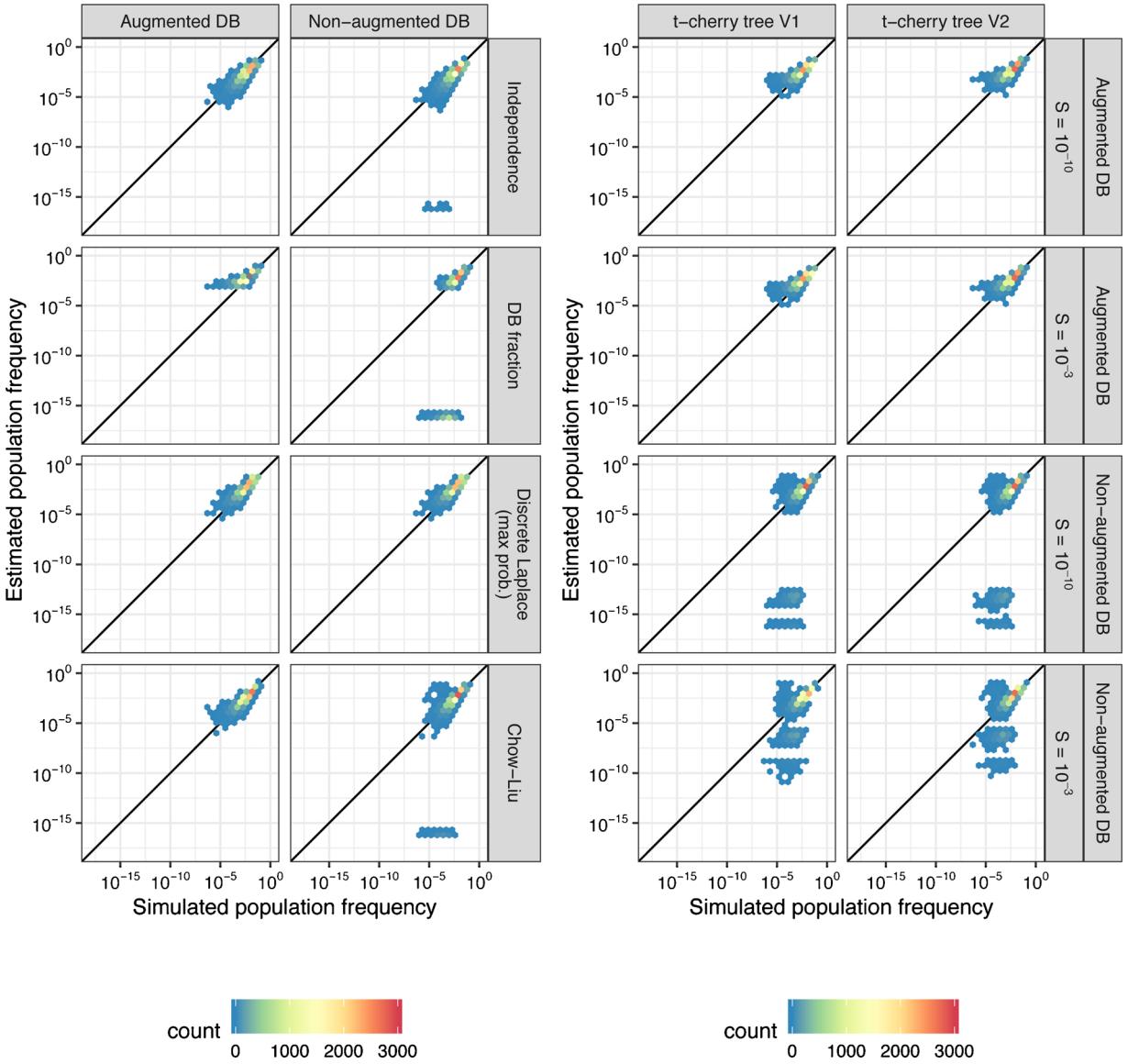


Fig. 7. Effect of augmenting the database for the 4-LMS. The superimposed identity line ($y = x$) is shown. For this figure the results from databases of size 500 and 1000 were merged. Estimated frequencies below 10^{-16} were replaced by 10^{-16} for the non-augmented databases. DB: database, S: smoothing parameter, V1: first version for the t-cherry tree construction (from the Chow–Liu tree), V2: second version for the t-cherry tree construction (direct construction).

not just from cliques). Although not subject of this study, it remains an interesting question which algorithms are best for approximating the optimal t-cherry tree within a reasonable running time and how they compare to the global optimum. It would also be of interest to go outside the domain of t-cherry trees, although it makes the problem of finding the best model even harder. One approach could be to remove some of the edges in the final t-cherry tree, i.e. if conditional independence can be established.

The introduced t-cherry tree method together with the compared methods are aimed at estimating Y haplotype frequencies. We want to stress the well-known fact that frequencies have to be distinguished from match probabilities. This is especially relevant for Y haplotype profiles. Brenner [36] pointed out that for Y profile evidence evaluation relatedness plays a crucial role. Therefore, it is usually not acceptable to equate Y haplotype frequencies with match probabilities in likelihood ratio calculations. Andersen and Balding [37] as well as Caliebe and Krawczak [11] stressed that the suspect population has to be considered where often a much higher degree of relatedness is present than in the database population from which the haplotype frequency is calculated. Andersen and Balding [37,38] proposed a new approach to report the evidential value of a Y haplotype match by considering the distribution of the number of males with a matching Y profile. Taylor et al. [10] take a similar stance in that they consider matching profiles to be most likely due to related individuals.

One limitation of this study is its reliance on the suitability of the population simulation models employed. Indeed, only the single-step symmetric mutation model was implemented, which is known to be an oversimplification [39–41]. A replacement by more realistic models would be a worthwhile future project. Another aspect which might be worth including is migration. Note that our population simulations only apply to large populations with mating, birth rate and growth structure as used in our models. Further research is necessary for small isolated populations, or populations with strong bottlenecks, or under strong selection. The same applies to inbred populations. Nevertheless, our results are consistent over various scenarios covering different growth behaviour and reproductive success. The t-cherry tree estimation method showed the best performance in the set of compared methods. Whereas the well-established discrete Laplace method has a similar performance to the Chow–Liu method, the t-cherry tree approach has lower bias and mean squared error than both methods when comparing

Table 5
Comparison of augmented and non-augmented estimators.

DB size	Estimator	Augmented		Non-augmented		
		MSE	Bias	MSE	Bias	n_0
500	DB fraction	0.17	0.21	36.1	−2.78	1086
500	Independence	0.11	−0.11	0.79	−0.19	24
500	Chow–Liu	0.06	0.00	4.33	−0.39	138
500	t-cherry tree V1, $S = 10^{-3}$	0.11	0.11	1.34	−0.36	0
500	t-cherry tree V1, $S = 10^{-10}$	0.11	0.11	11.5	−1.12	75
500	t-cherry tree V2, $S = 10^{-3}$	0.11	0.11	1.33	−0.37	0
500	t-cherry tree V2, $S = 10^{-10}$	0.11	0.11	11.7	−1.14	69
500	Discrete Laplace	0.10	0.15	0.09	0.13	0
1000	DB fraction	0.11	0.13	21.2	−1.67	659
1000	Independence	0.12	−0.12	0.35	−0.15	8
1000	Chow–Liu	0.05	−0.02	1.89	−0.20	63
1000	t-cherry tree V1, $S = 10^{-3}$	0.07	0.05	0.71	−0.21	0
1000	t-cherry tree V1, $S = 10^{-10}$	0.07	0.05	5.05	−0.62	37
1000	t-cherry tree V2, $S = 10^{-3}$	0.07	0.05	0.73	−0.22	0
1000	t-cherry tree V2, $S = 10^{-10}$	0.07	0.05	6.40	−0.65	32
1000	Discrete Laplace	0.09	0.14	0.09	0.13	0

The mean squared errors (MSEs) and biases are given on $\log_{10}$ scale. A positive (negative) sign of the bias shows on overestimation (underestimation) of the simulated frequencies.
DB: database, V1: first version for the t-cherry tree construction (from the Chow–Liu tree), V2: second version for the t-cherry tree construction (direct construction), n_0: number of haplotypes with an estimated frequency of 0, S: smoothing parameter.

estimated and simulated haplotype frequencies. Moreover, the t-cherry estimate is easier and quicker to calculate than the discrete Laplace estimate. When it comes to Y haplotype frequency estimation, the t-cherry tree method therefore shows promising potential.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.fsigen.2019.102214.

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